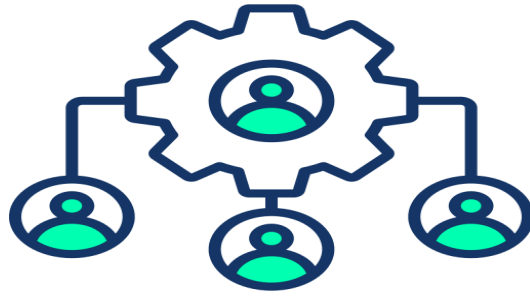




Federal Urdu University
OF ARTS, SCIENCE AND TECHNOLOGY



FUFAST

ASSIGNMENT TOPIC:
SMART STUDENT PORTAL AI AGENT

Name: Ayesha Raza

D/O: Raza Ullah Khan

Section: 6B

COURSE: AI

Professor: Ma'am Uzma Afzal

Due date 6-sep-2026

1. Identify your problem in the context of agent.

University students have to repeatedly check their grades, attendance, upcoming deadlines, and fee status on their own, which is time taking and often leads to missed deadlines, unnoticed low attendance, or forgotten fee due dates simply because no automatic system alerts them in time.

To solve this problem, a Smart Student Portal AI Agent has been designed that works on predefined condition-action rules. This agent checks the current state of the portal and reacts immediately based on a set of fixed rules

For example, sending a warning if attendance drops, or sending a reminder if a deadline is approaching.

2. Problem formulation.

Textual Representation

Sensor:

The agent senses the current attendance percentage, latest grade updates, upcoming deadline dates, fee due dates, and any query submitted by the student through the portal.

State:

The state represents the current condition of the student's data on the portal such as attendance level, latest grades, nearest deadline, and fee status as sensed at that moment, without depending on any previous state.

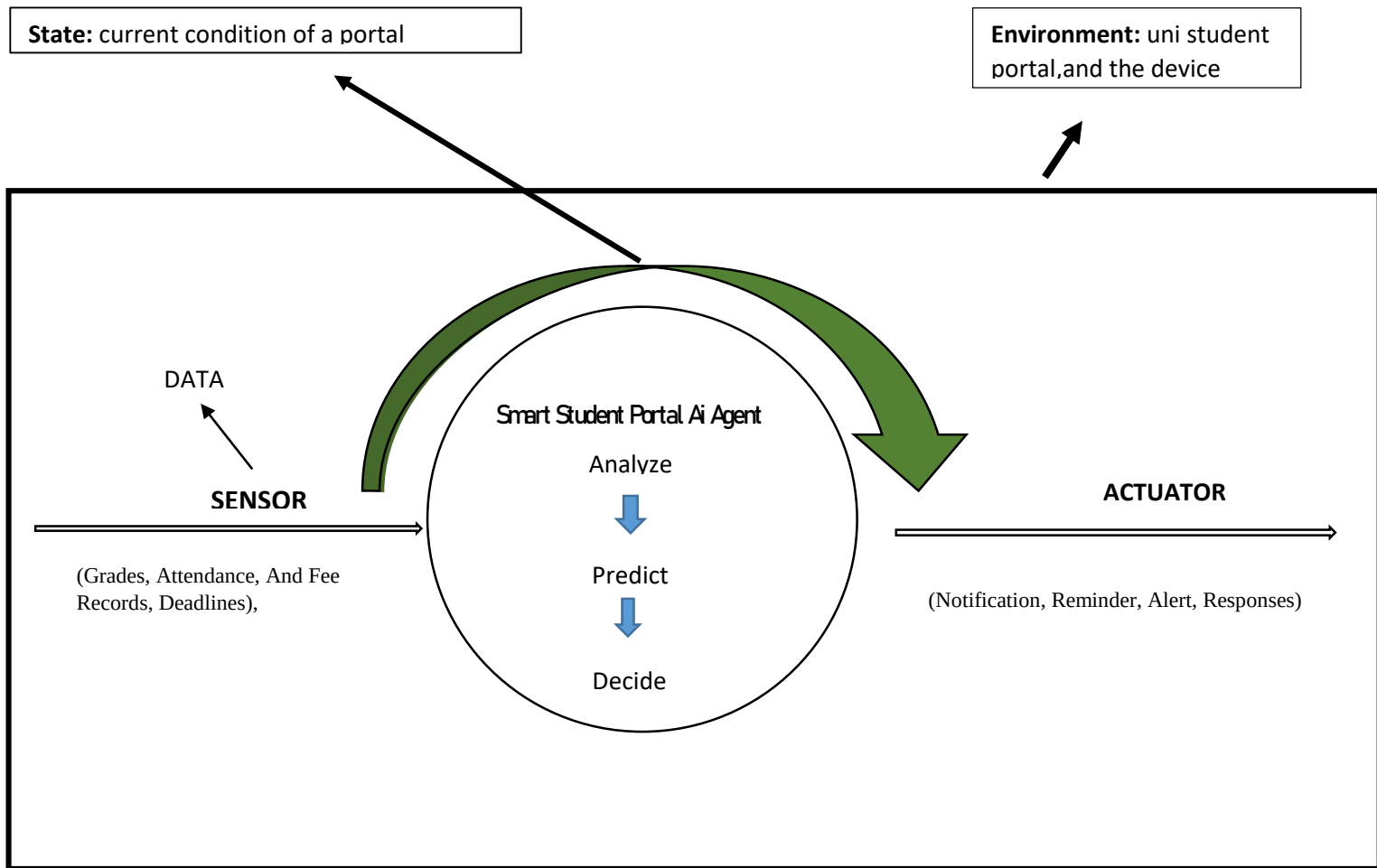
Actuator:

The agent acts by sending notifications or reminders, displaying information (grades deadlines) on the dashboard, and giving a reply to a student query.

Environment:

The environment consists of the university's online student portal, (grades, attendance, and fee records), the deadline/course calendar, and the student's device through which alerts and information are delivered.

GRAPHICAL PRESENTATION



3. Evaluate in the Context of PEAS.

Performance Measure:

The success of the agent is measured by how accurately and quickly it detects a triggering condition (like low attendance, an approaching deadline, or a due fee) and sends the correct alert. Other measures include timeliness of notifications, correctness of grade attendance information shown, and overall student satisfaction with getting timely alerts.

Environment: The environment is the university's online student portal, including its databases for grades, attendance, and fee records, the deadline/course calendar, and the student's device (browser or mobile app) where the alerts and information are received.

Actuators: The agent acts through notifications and reminders displaying grade or deadline updates on the dashboard, and giving replies to student queries.

Sensors: The agent senses the current attendance percentage, newly posted grades, upcoming deadline dates, fee due dates, and any query submitted by the student through the portal.

4. Identify the Environmental Properties in the Context of Agent.

• **Partially Observable:** The agent does not know everything about the student. It mainly observes available information such as attendance, grades, deadlines, and fees.

• **Dynamic:** The environment can change at any time. For example, a new grade may be uploaded, attendance may change, or a deadline may approach.

• **Sequential:** Current events can affect future decisions. For example, low attendance today may lead to an attendance warning later.

• **Discrete:** Many events occur as separate actions, such as a grade being posted, a fee becoming due, or a student submitting a query.

• **Deterministic:** For known portal data, the same input normally produces the same result. For example, if attendance is below the required percentage, the agent can generate an attendance warning.

• **Multi-Agent:** The environment involves multiple participants, including students, teachers, administrators, and the AI agent.